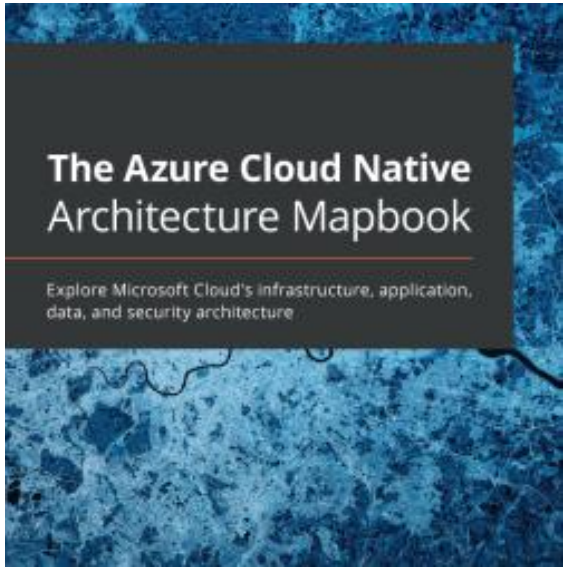
From outages to always on

Infra & application-level tips

Me in a nutshell

- 26 years+ in IT industry
- Swiss Knife Cloud Architect (Azure MVP)
- K8s SME
- AI enthusiast
- Author



The Azure Cloud Native Architecture Mapbook

"The Azure Cloud Native Architecture Mapbook by Stéphane Eyskens is a different kind of book in the Azure genre. It's not your regular "how to deploy this service for this problem" reading material. It truly focuses on developing an architectural mindset. I really liked the "map" approach used by the author and how it is followed in each chapter, giving the reader a clear mental model of Azure domains. The chapters are structured in a progressive way, moving from foundational concepts to advanced topics. Stéphane has done an excellent job of keeping it very realistic, as the book talks about organizational constraints, functional requirements, and much more.

There could be a very small improvement made by adding end-of-chapter takeaways. It's less about "how to build X" and more about "how to think clearly when building on Azure," which is exactly what makes it so valuable."

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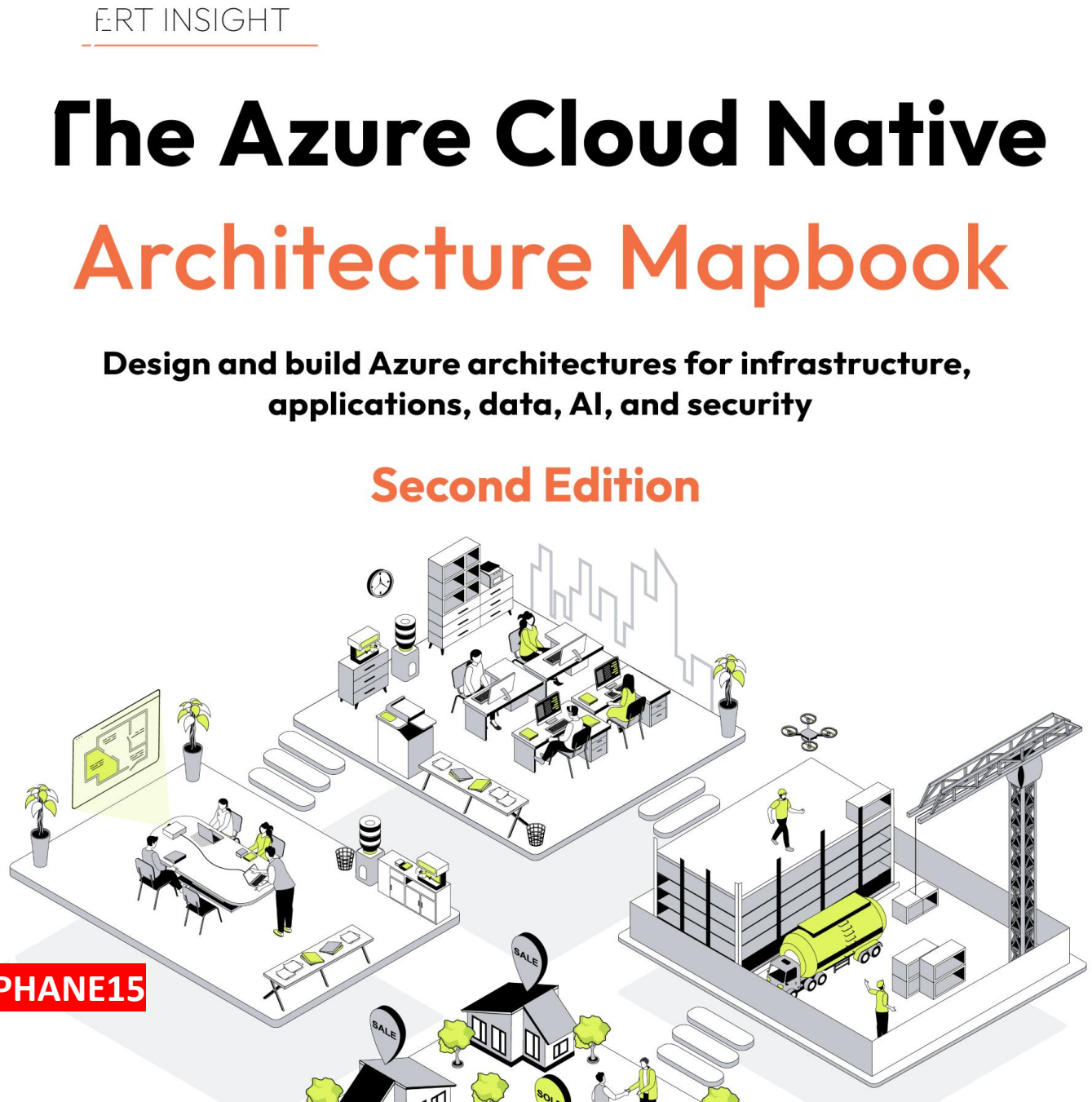


Types of failures and possible mitigations

Application-level (e.g. memory leaks, resource exhaustion, etc.)	Multiple instances
Hardware	Availability Sets
Data Center	Availability Zones
Regional Outages	Multi-region architectures

Problem Statement

Azure regions do not provide uniform service availability or multi-region capabilities, making it difficult for organizations to design and operate a consistent and reliable resilience strategy across different regional models.

Region Models and why it matters?

Paired Regions

Paired Regions are part of the same Geography (ie: West Europe and North Europe) and ensure Business Continuity and Disaster Recovery by exposing similar services and capabilities, as well as having Microsoft-level failover processes

Sovereign Regions

Sovereign Regions such as Azure US Government, Azure China respond to stricter regulations and targeting specific markets

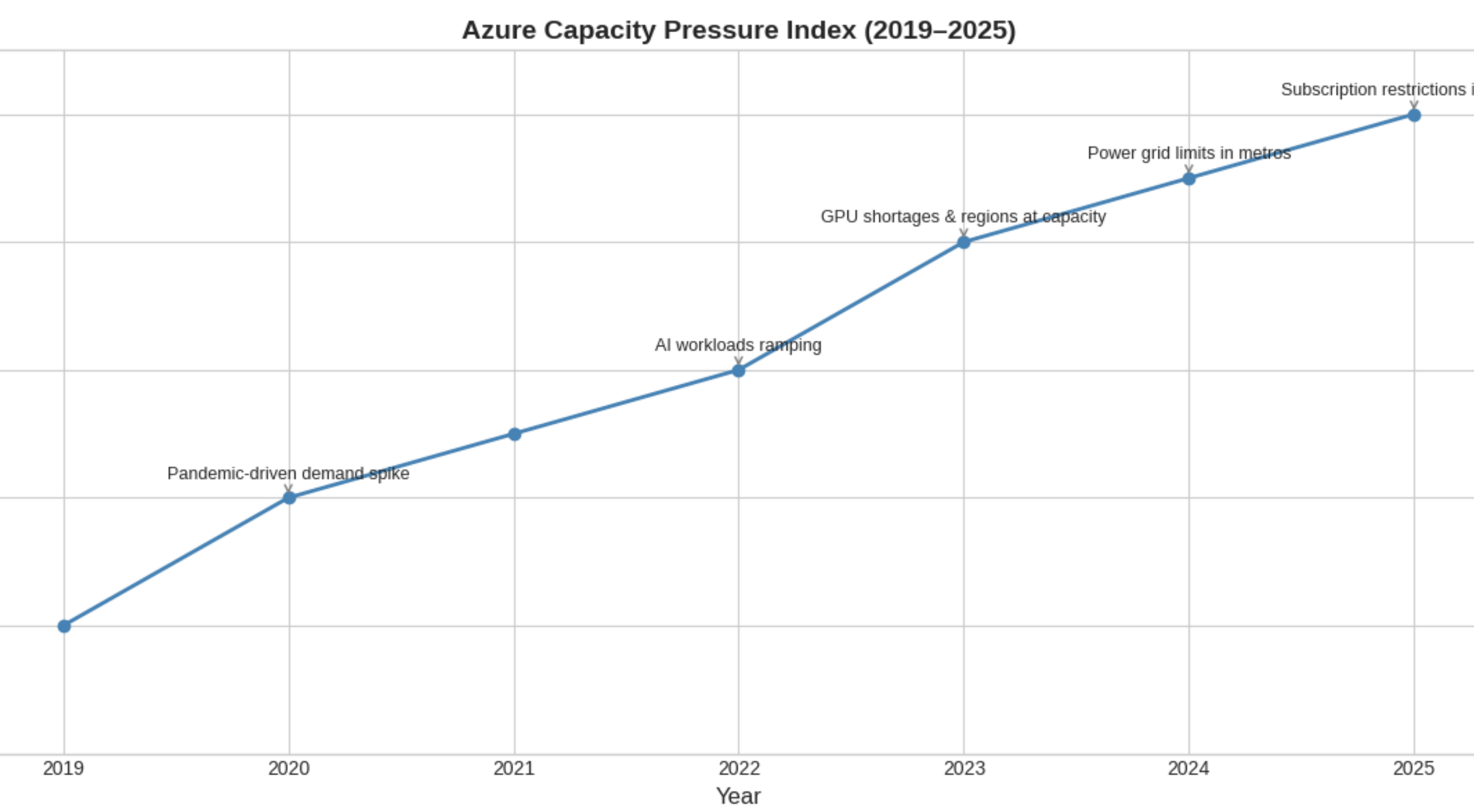
Restricted Regions

Restricted Regions such as France South are typically the pair of a non-restricted region (eg: France Central) and can be accessed only by specific customers

Local Regions

Local regions such as Azure Belgium Center (ABC) respond to a need for national data centers to accommodate data residency requirements. They do not support features such as GRS

Australia Central > AUSTRALIACENTRAL	Australia	Canberra	-	Australia Central 2	2018
Australia East > AUSTRALIAEAST	Australia	New South Wales	3	Australia Southeast	2014
Australia Southeast > AUSTRALIASOUTHEAST	Australia	Victoria	-	Australia East	2014
Austria East > AUSTRIAEAST	Austria	Vienna	3	-	2025
Belgium Central > BELGIUMCENTRAL	Belgium	Brussels	3	-	2025
Brazil South > BRAZILSOUTH	Brazil	Sao Paulo State	3	South Central US	2014
Canada Central > CANADACENTRAL	Canada	Toronto	3	Canada East	2016
Canada East > CANADAEAST	Canada	Quebec	-	Canada Central	2016
Central India > CENTRALINDIA	India	Pune	3	South India	2015
Central US >	United States	Iowa	3	East US 2	2014


Azure Capacity Pressure Index (2019–2025)
Subscription restrictions i
Power grid limits in metros
GPU shortages & regions at capacity
AI workloads ramping
Pandemic-driven demand spike
2019
2020
2021
2022
2023
2024
2025
Year

Why should you even envision multi-region?

Large Power Grid Failures

Remember the Iberian Peninsula blackout?

Natural Disasters

- *Floods*
- *Earthquakes*
- *Wildfires*
- *Storms*

Others

- *Fiber cut*
- *Local cloud breakout down*
- *Regulatory requirements*
- *Bombing…*

How to build resilient architectures?

Some regions cannot do what others can.
Your continuity strategy must adapt

Infrastructure-level building blocks

Databases

Active Geo Replication

Failover Groups

Multi-region writes

Message and Event Brokers

- *Geo Disaster Recovery*
- *Active Geo Replication*

Storage

- *GRS, GZRS, RA-GRS, RA-GZRS*
- *Object Replication*

Global Services

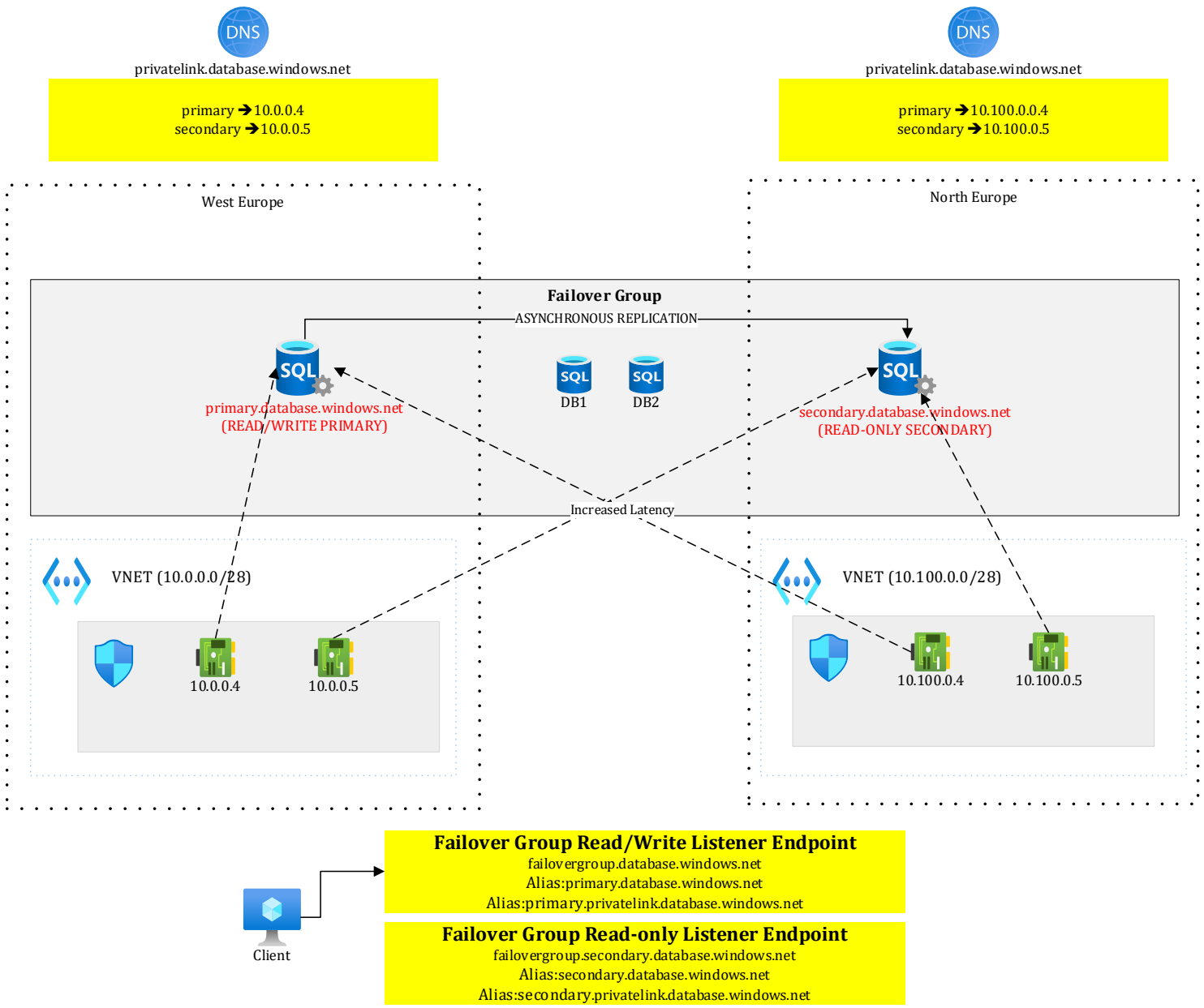
- *Traffic Manager*
- *Front Door*
- *DNS Zones, etc.*

Azure SQL

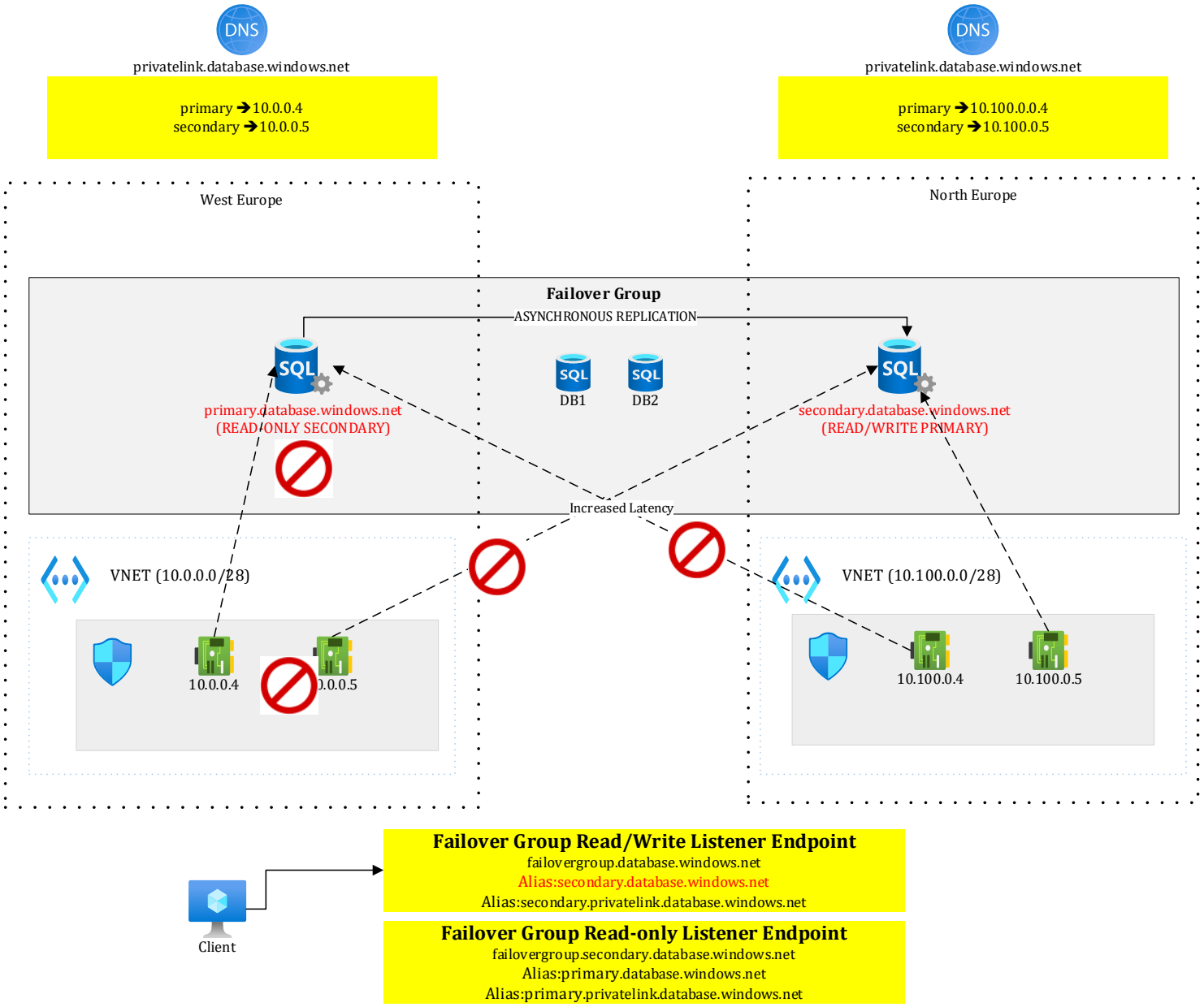
Failover Groups (Azure SQL+MI)

- Customer or Microsoft-Managed Failover (policy)
- One or more databases per group
- Active/Active (read-only replicas)
- Single listener endpoint (no impact on client)

Failover Groups BEFORE Failover



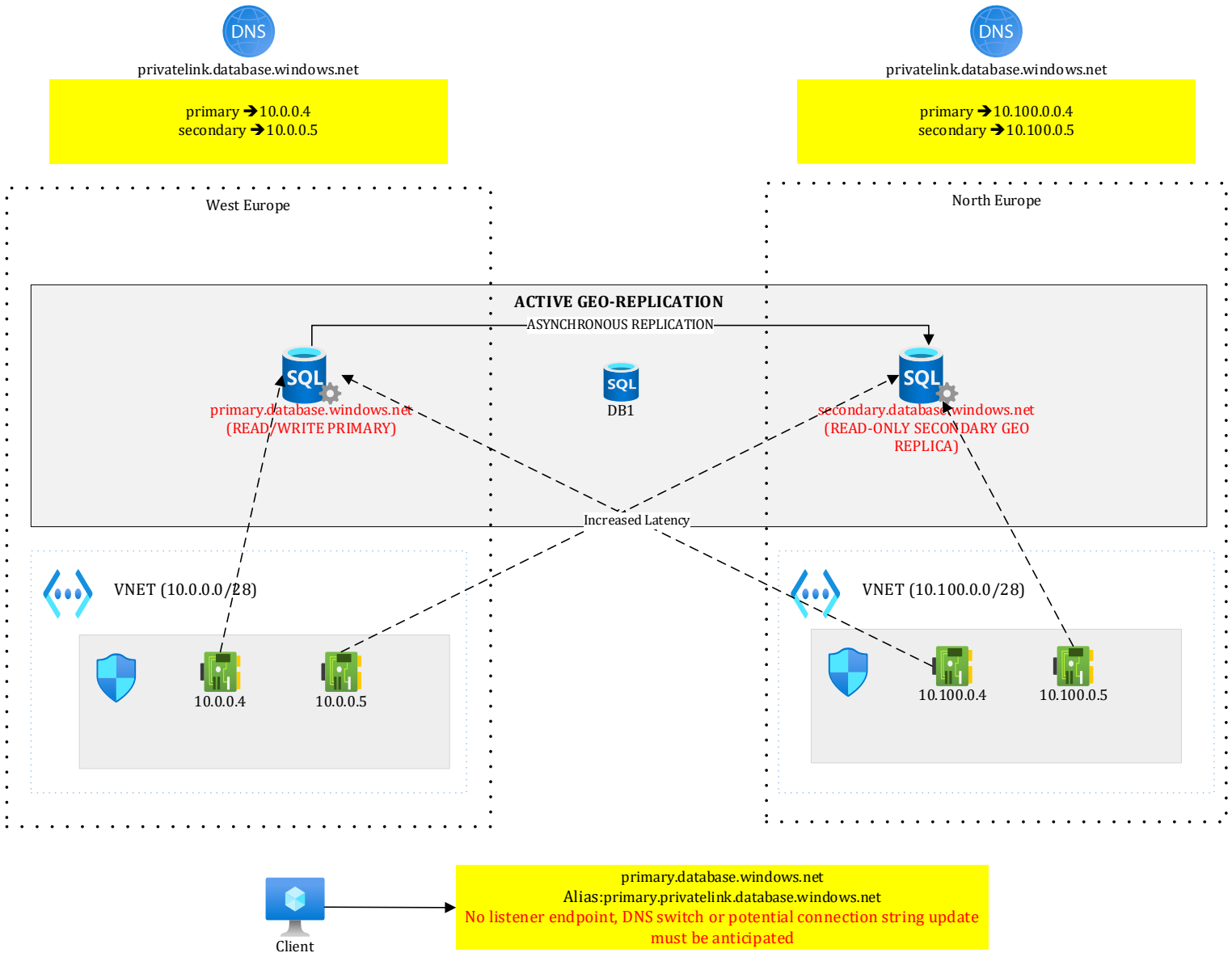
Failover Groups AFTER Failover



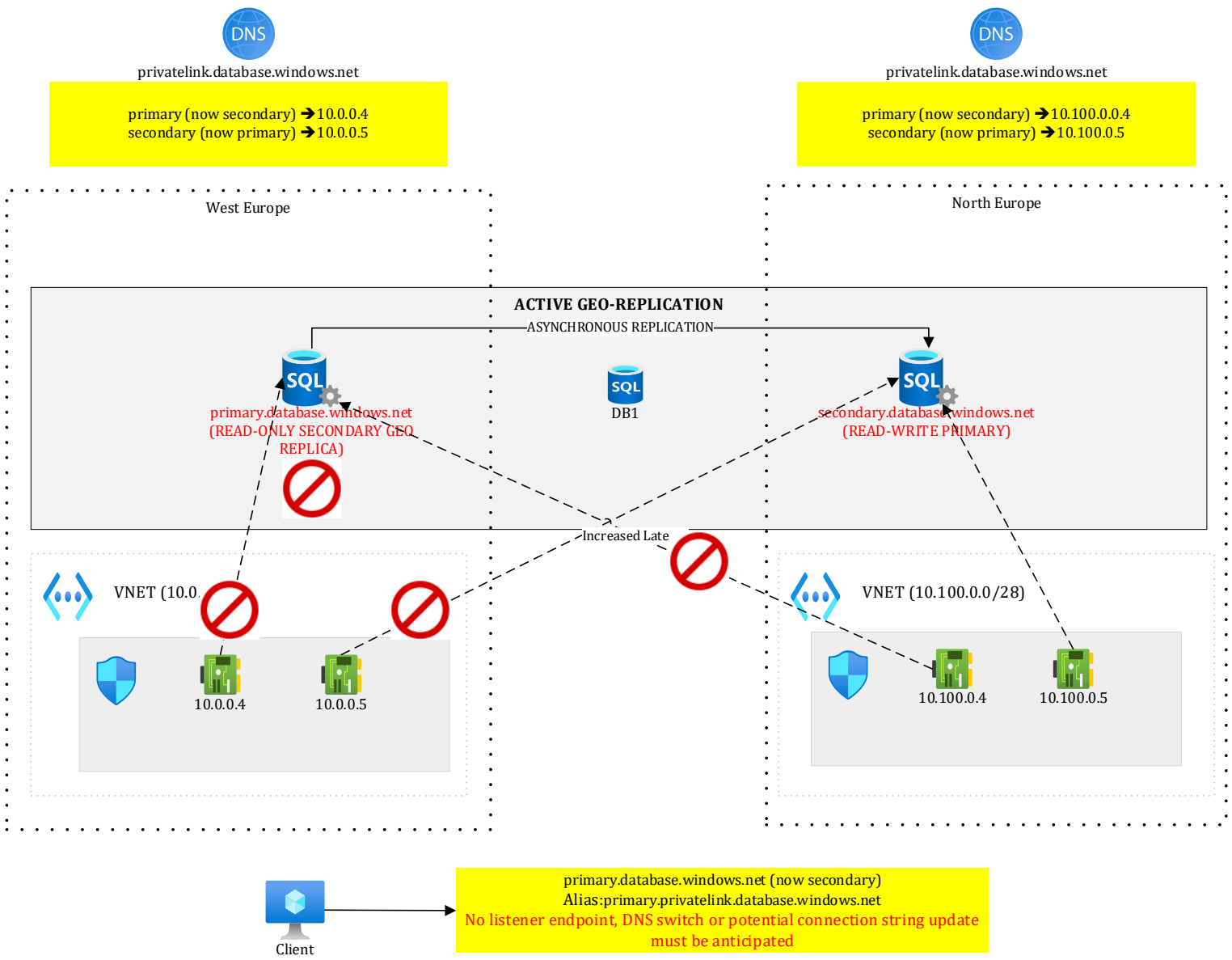
Active Geo-Replication

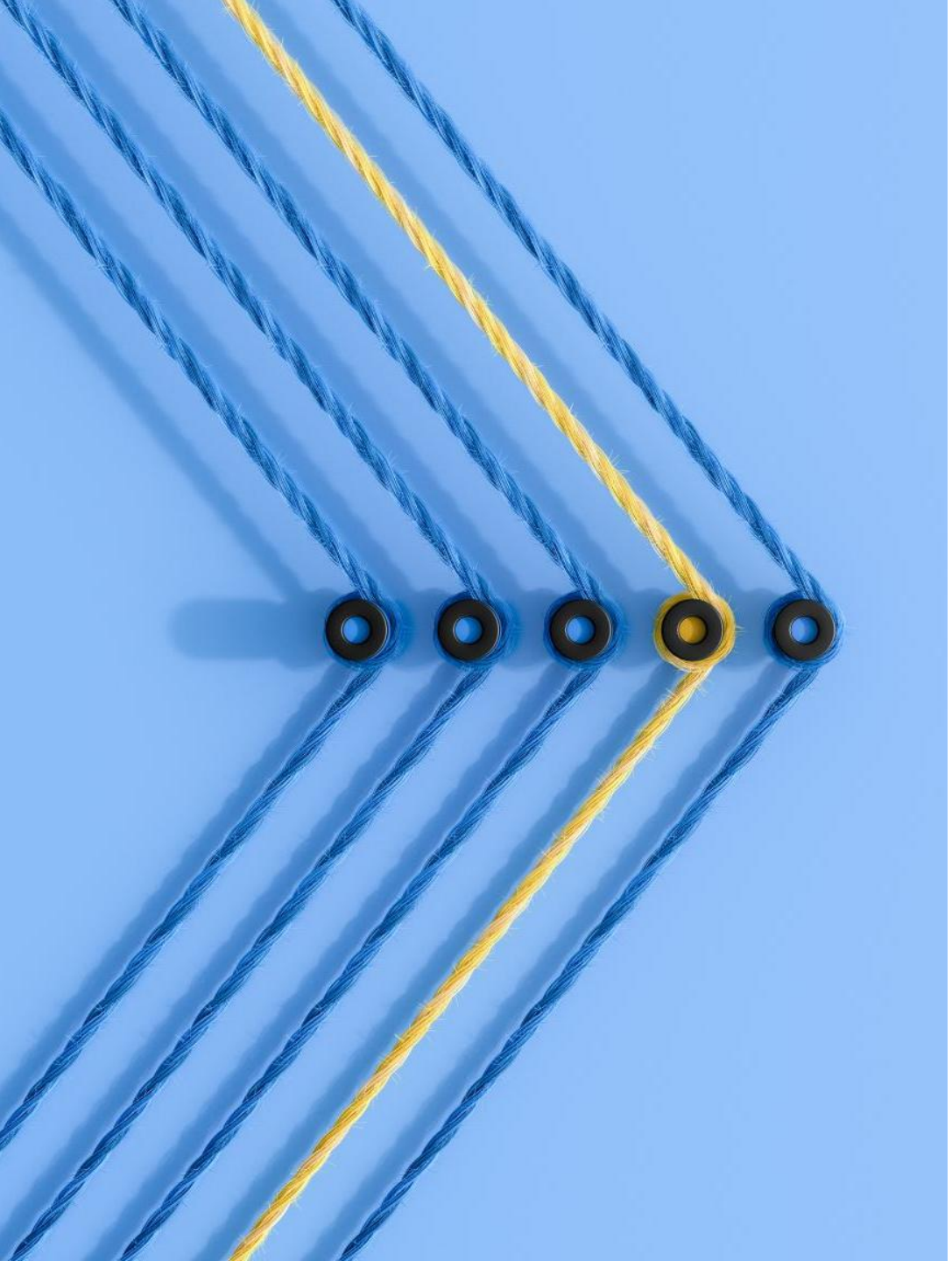
- Database-level feature
- Read replicas can be in the same or different regions
- Data replication is asynchronous
  - sys.dm_geo_replication_link_status contains information about replication lag and replication timestamps as well as replication state (pending, seeding, catch-up, suspended)

Active Geo Replication BEFORE failover



Active Geo Replication AFTER failover



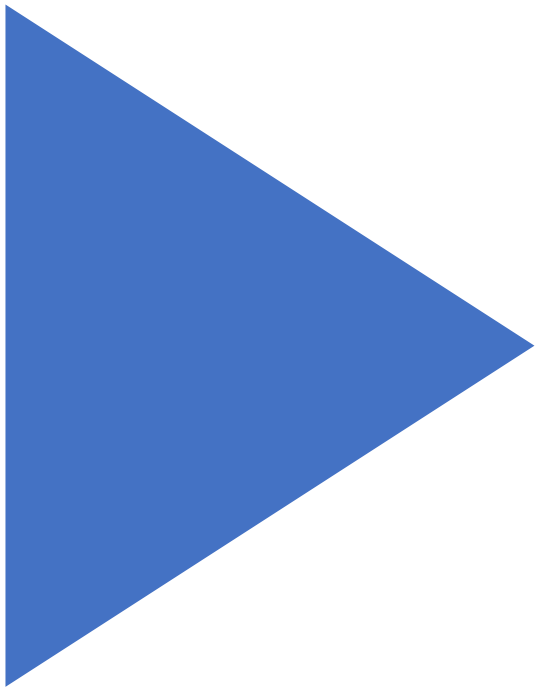


Application-level tricks

- Have both connection strings as part of the application configuration and implement fallback mechanisms to switch from primary to secondary and vice versa
- Make your deployments multi-region aware and deploy the right connection string according to the region the workload is deployed to

Azure SQL+MI - Failover Groups vs Active Geo-Replication

Feature	Failover Groups	Active Geo-Replication
Scope	Group of databases	Individual databases (database level)
Failover Type	Automatic or manual	Manual only
Listener Endpoints	Stable (read/write & read-only)	Must update after failover
Secondary Replicas	One per group	Up to four per database
Read Workload Offloading	Yes (via read-only listener)	Yes
Azure SQL Managed Instance Support	✅ Yes	❌ No



Wanna play with it? (simplified setup)

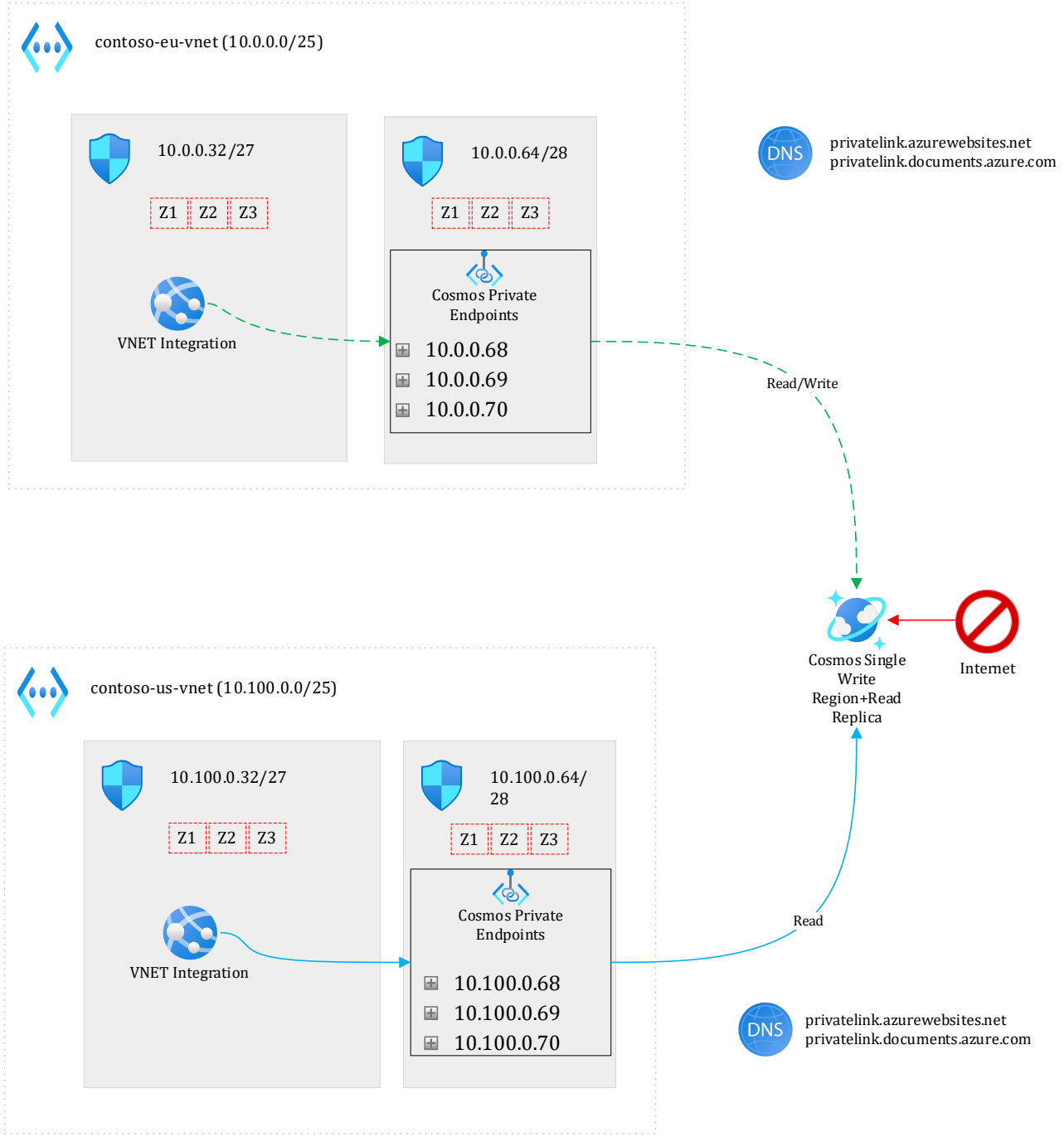
https://github.com/stephaneey/azure-and-k8s-architecture/tree/main/availability-samples/sql

Cosmos DB

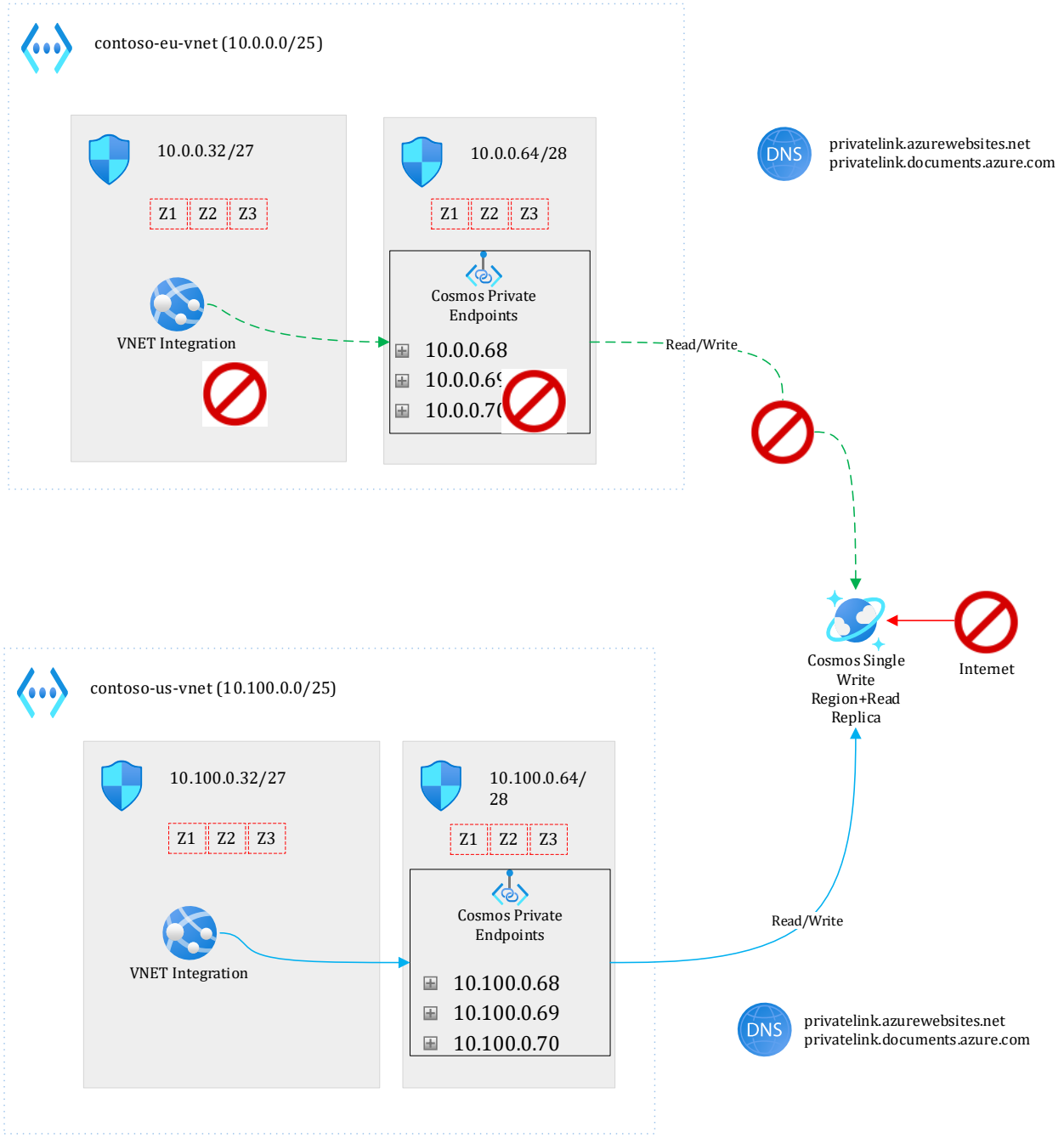
Features

- As of 03/2026, the only OLTP store that supports multi-region writes
- Can be used for active/active and active/passive scenarios

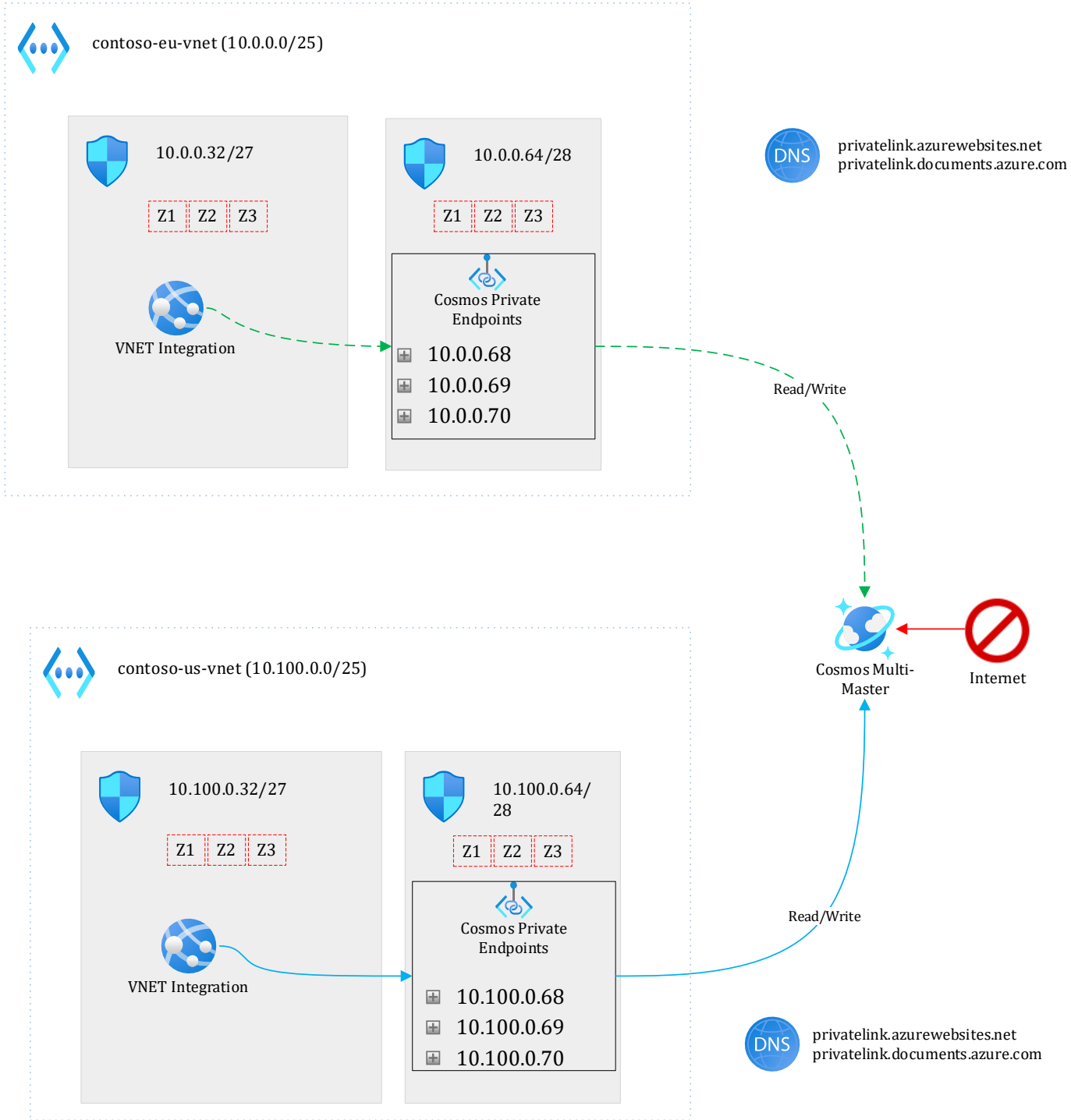
Single write - BEFORE regional outage



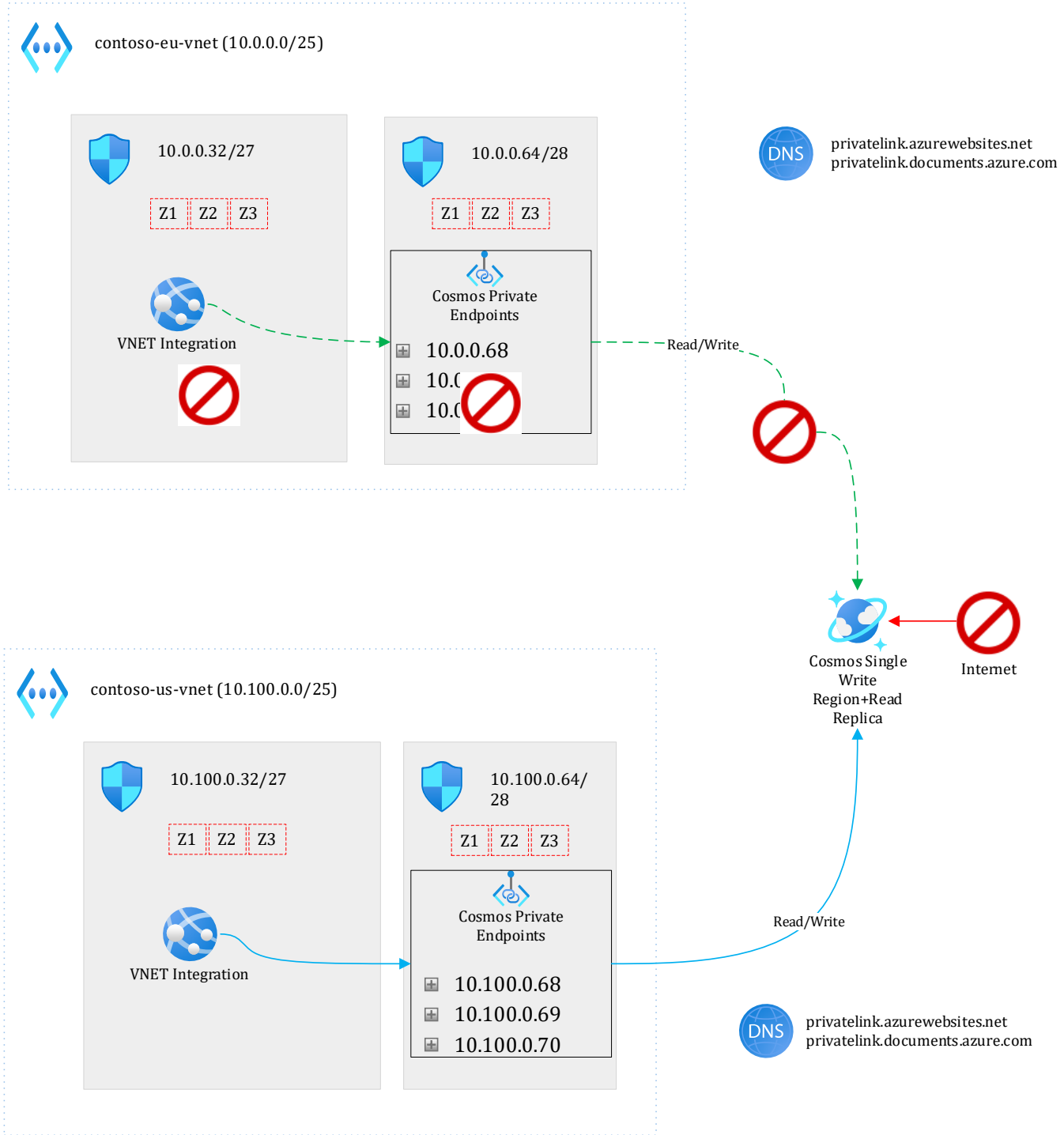
Single write - AFTER regional outage

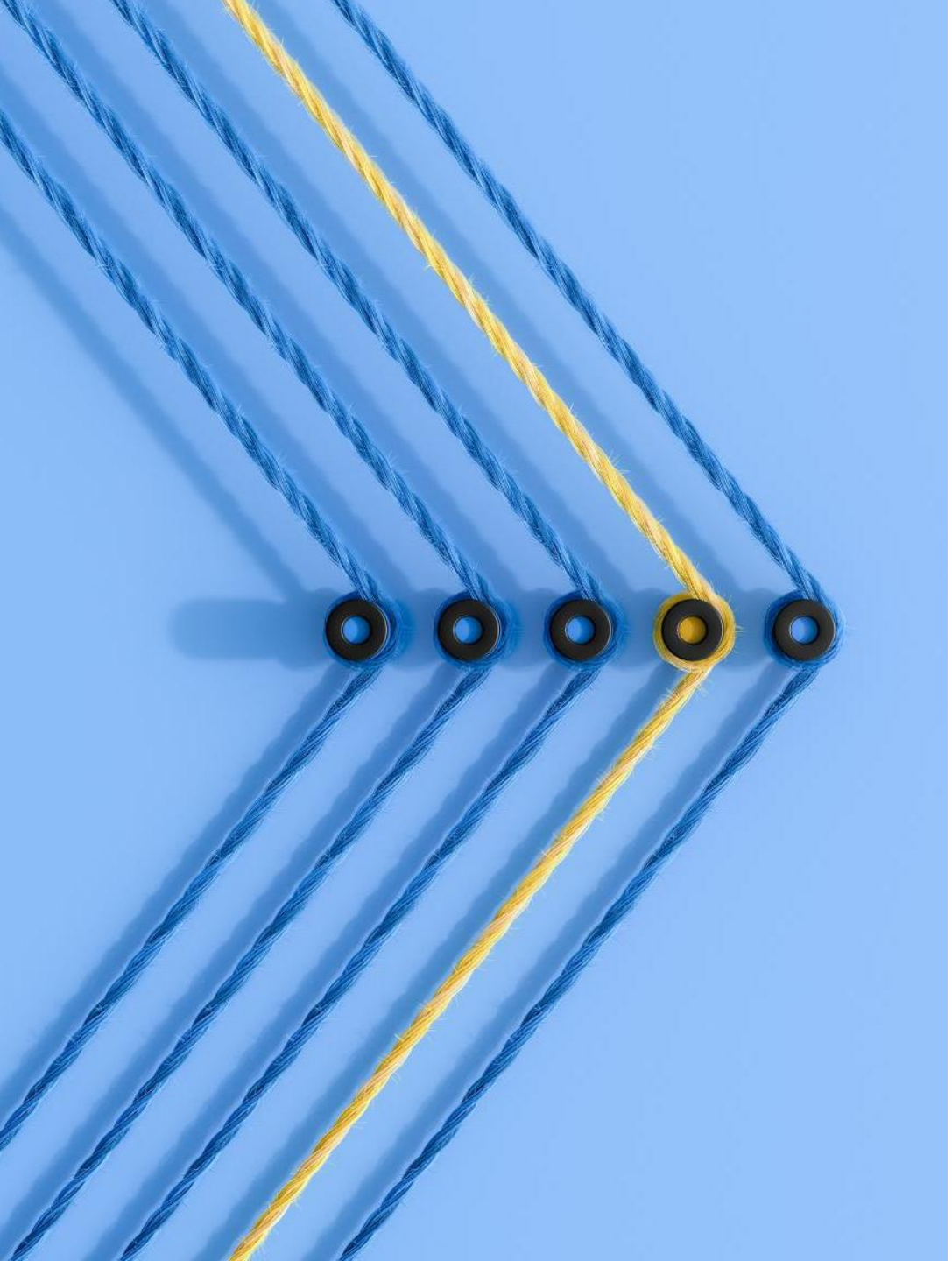


Multi-region writes - BEFORE regional outage



Multi-region writes - AFTER regional outage

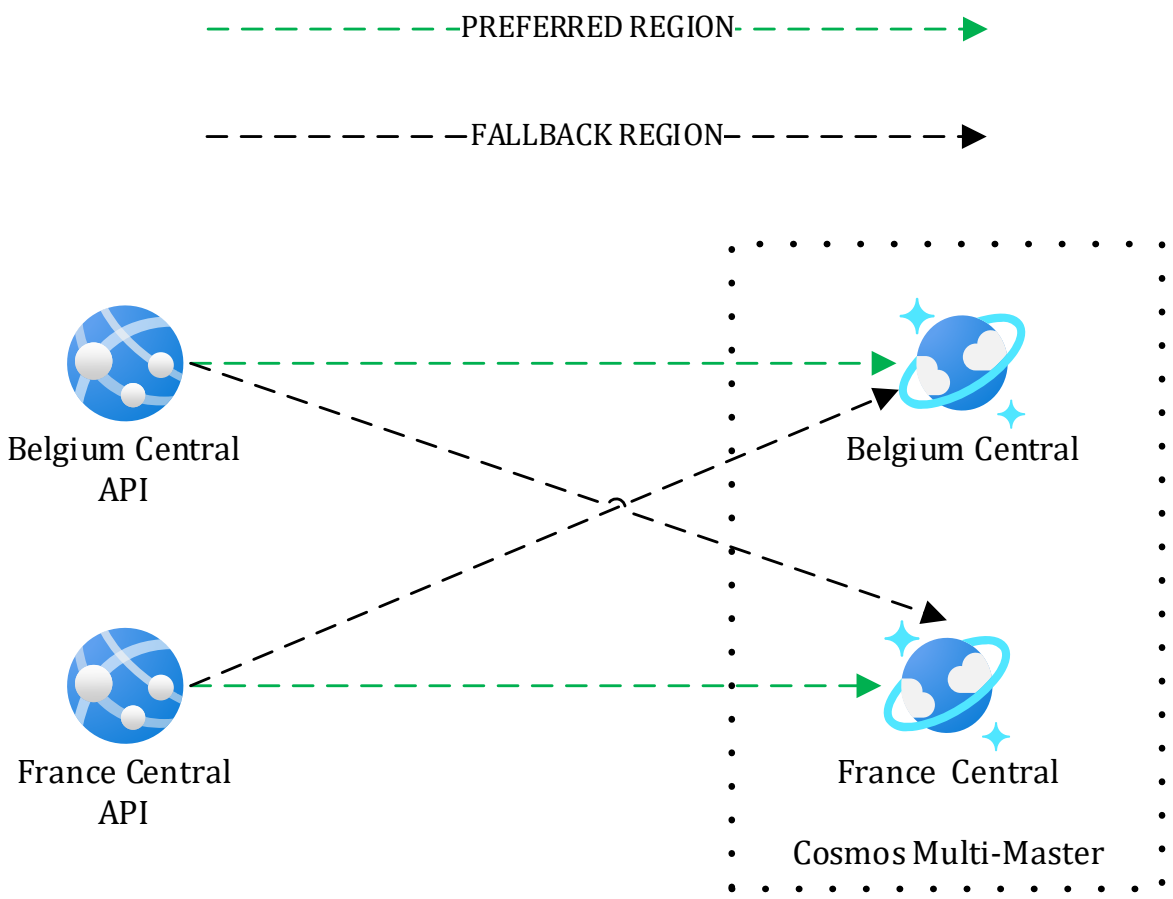




Application-level tricks

- The Cosmos DB SDK perfectly deals with multi-region scenarios.
- Define your preferred and fallback regions as part of the configuration and the SDK will do the rest.

Wanna play with it? (simplified setup)



https://github.com/stephaneey/azure-and-k8s-architecture/tree/main/availability-samples/cosmos

Azure Storage

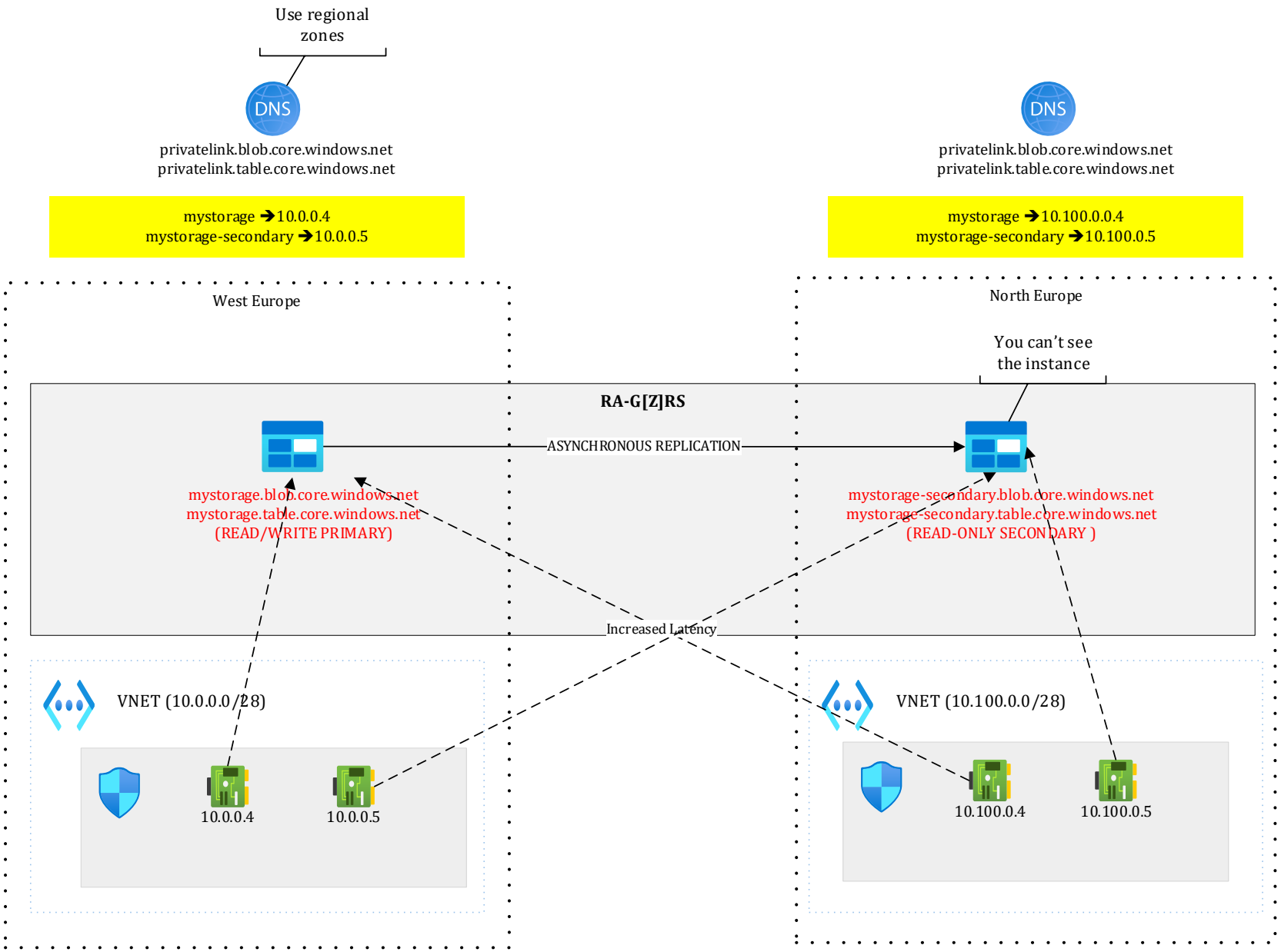
Features

- Remember that storage is not only about blobs, but also about queues, tables and files
- Storage Account is used by many other Azure services (e.g. Backup Vault & Recovery Site Vault, default state store for Durable Functions, etc.)
- GRS handles all sub-resources with some limitations for Azure Files
- Object Replication is currently limited to blobs and not even working with HNS

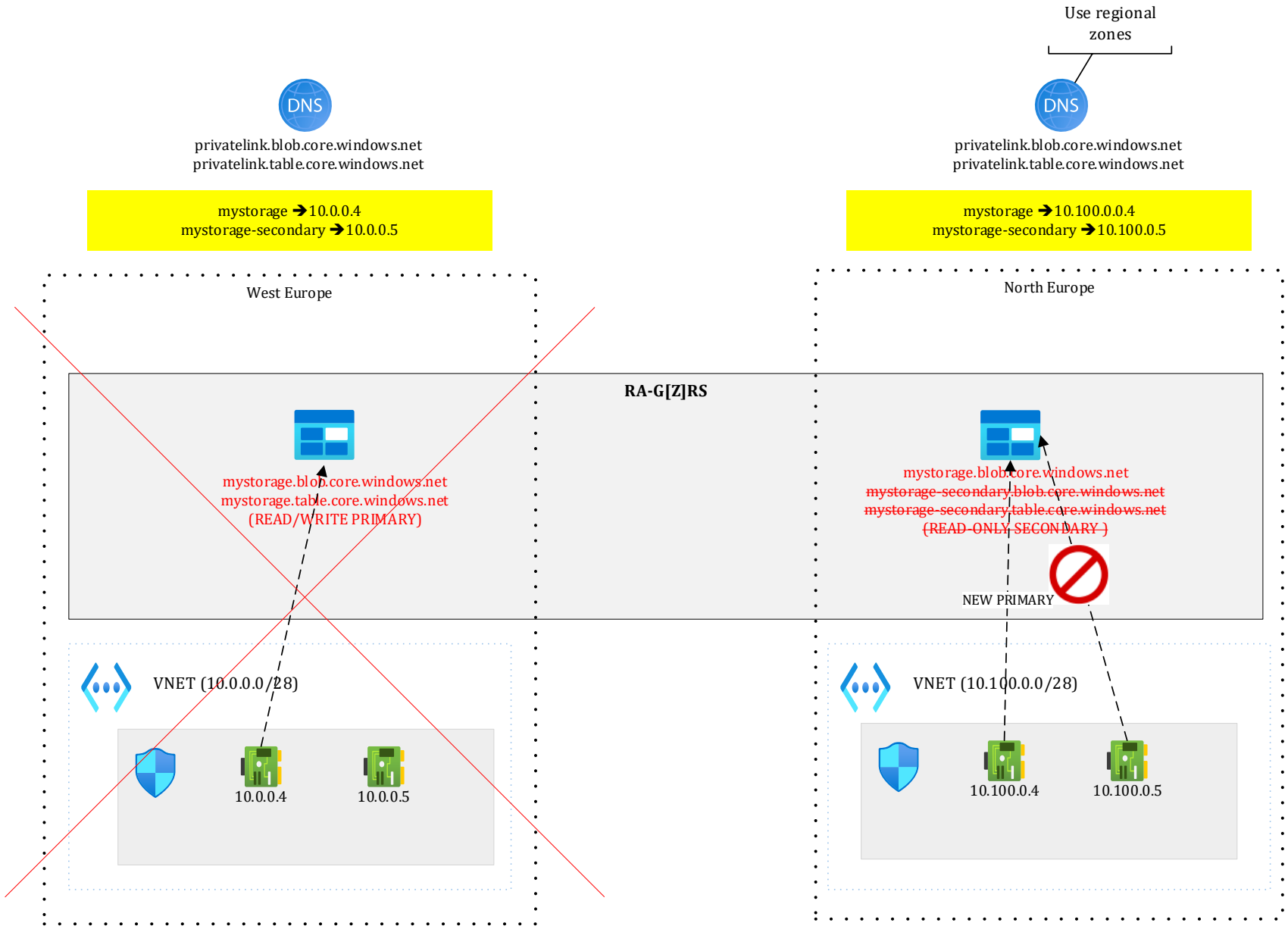
GRS challenges

- Still fully available in hero regions but such regions lack capacity.
- Still available for restricted regions but one cannot deploy compute to such regions…
- Not available at all in local regions

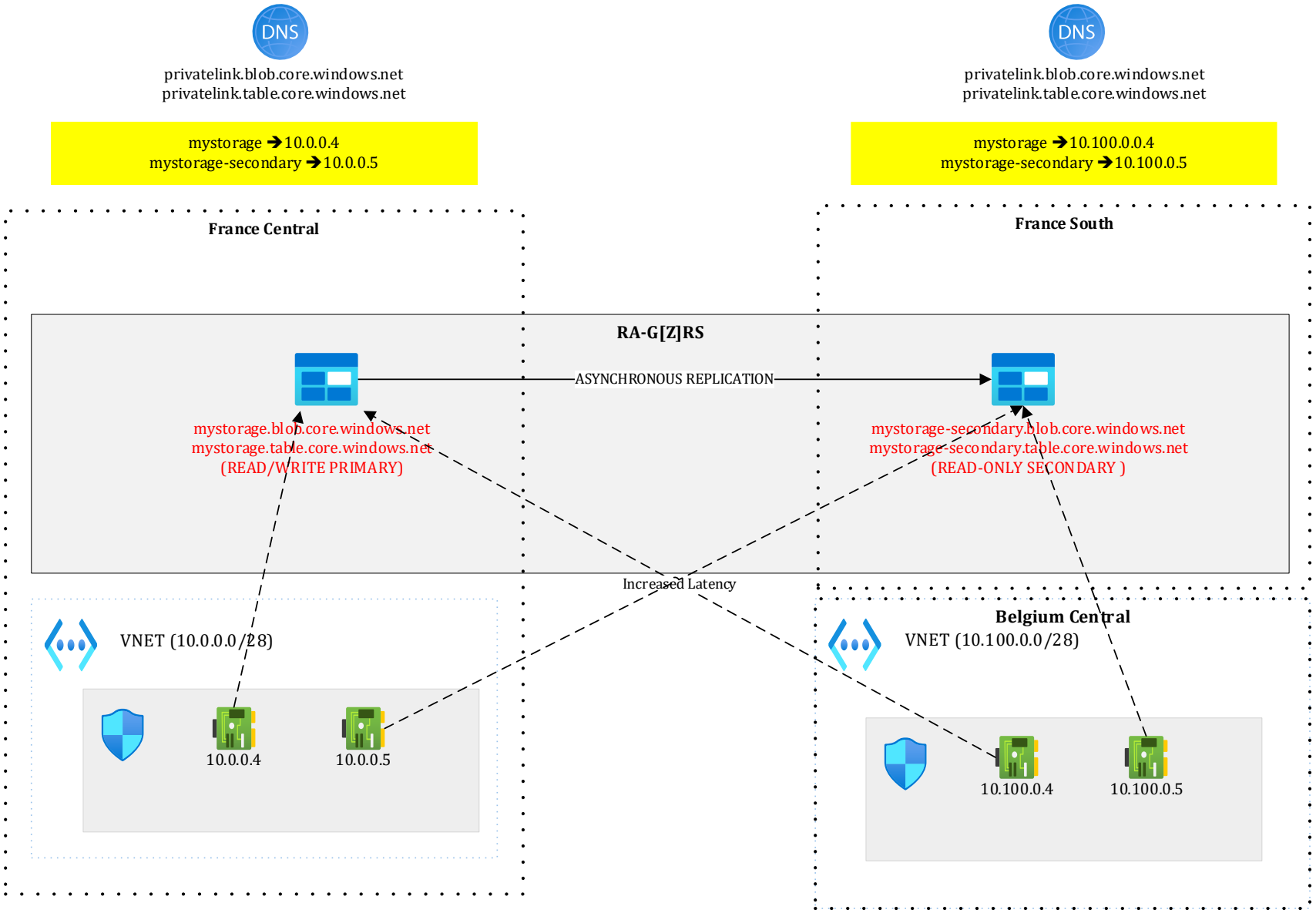
GRS hero regions – BEFORE failover



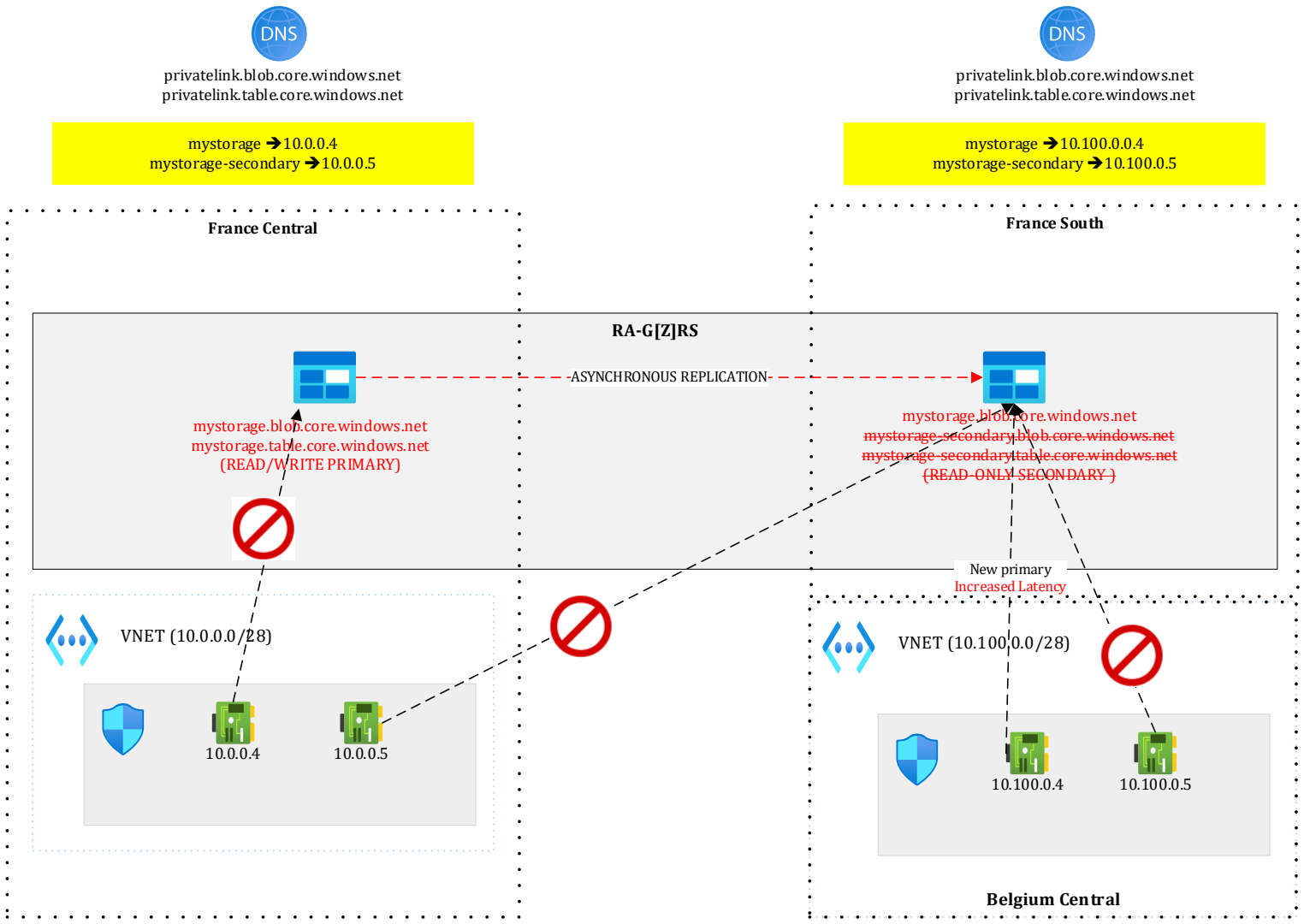
GRS hero regions – AFTER failover



GRS restricted regions – BEFORE failover



GRS restricted regions – AFTER failover



https://azure.microsoft.com/en-us/pricing/details/bandwidth/

€0.017 per GB

GRS – Things to consider

Geo Priority Replication

RPO <= 15 min

Forced vs Planned Failover

Forced Failover may incur data loss and switches the replication policy from GRS to LRS

Planned Failover makes sure primary and secondary are in sync and keeps GRS

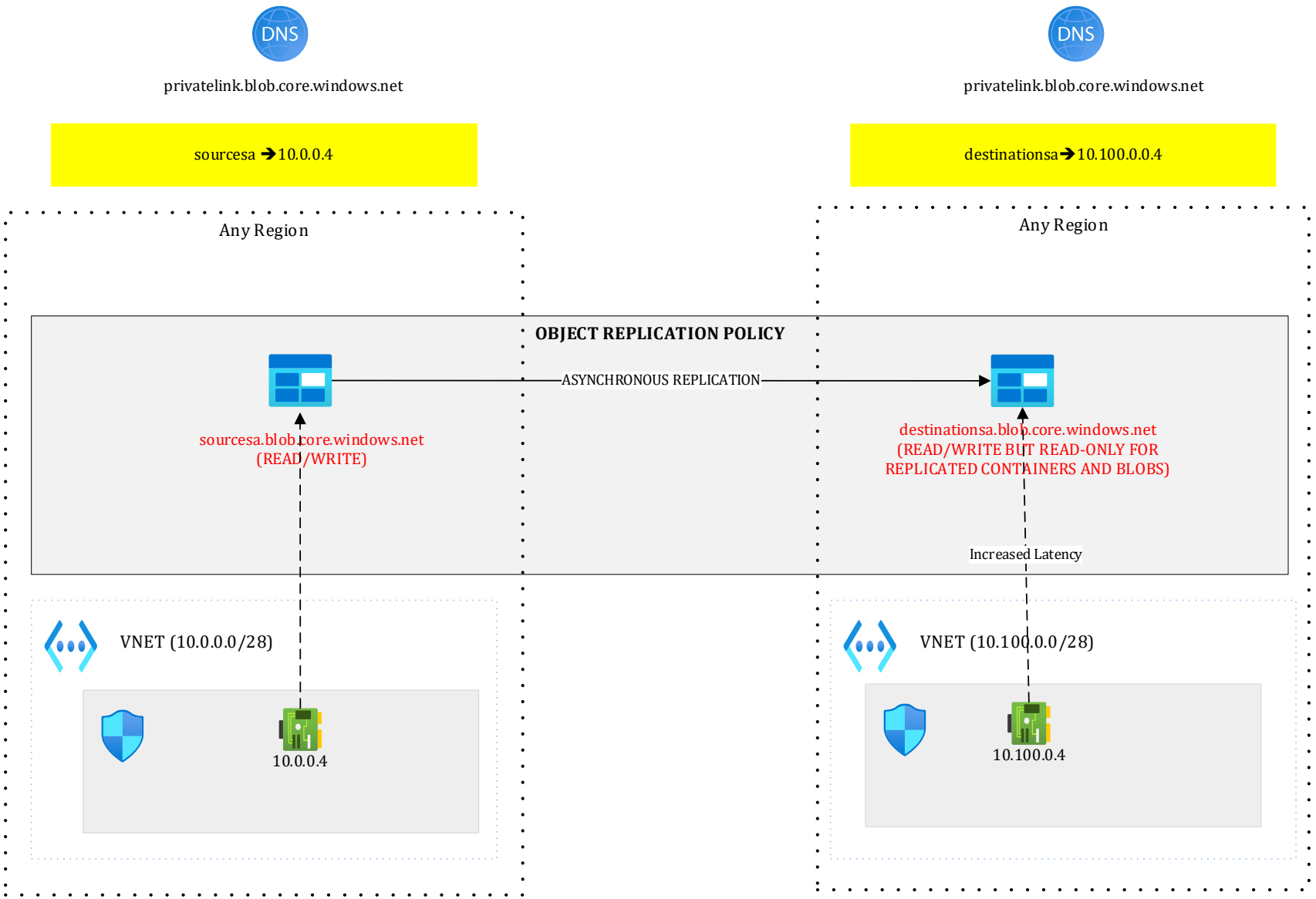
GRS Alternatives

Object Replication

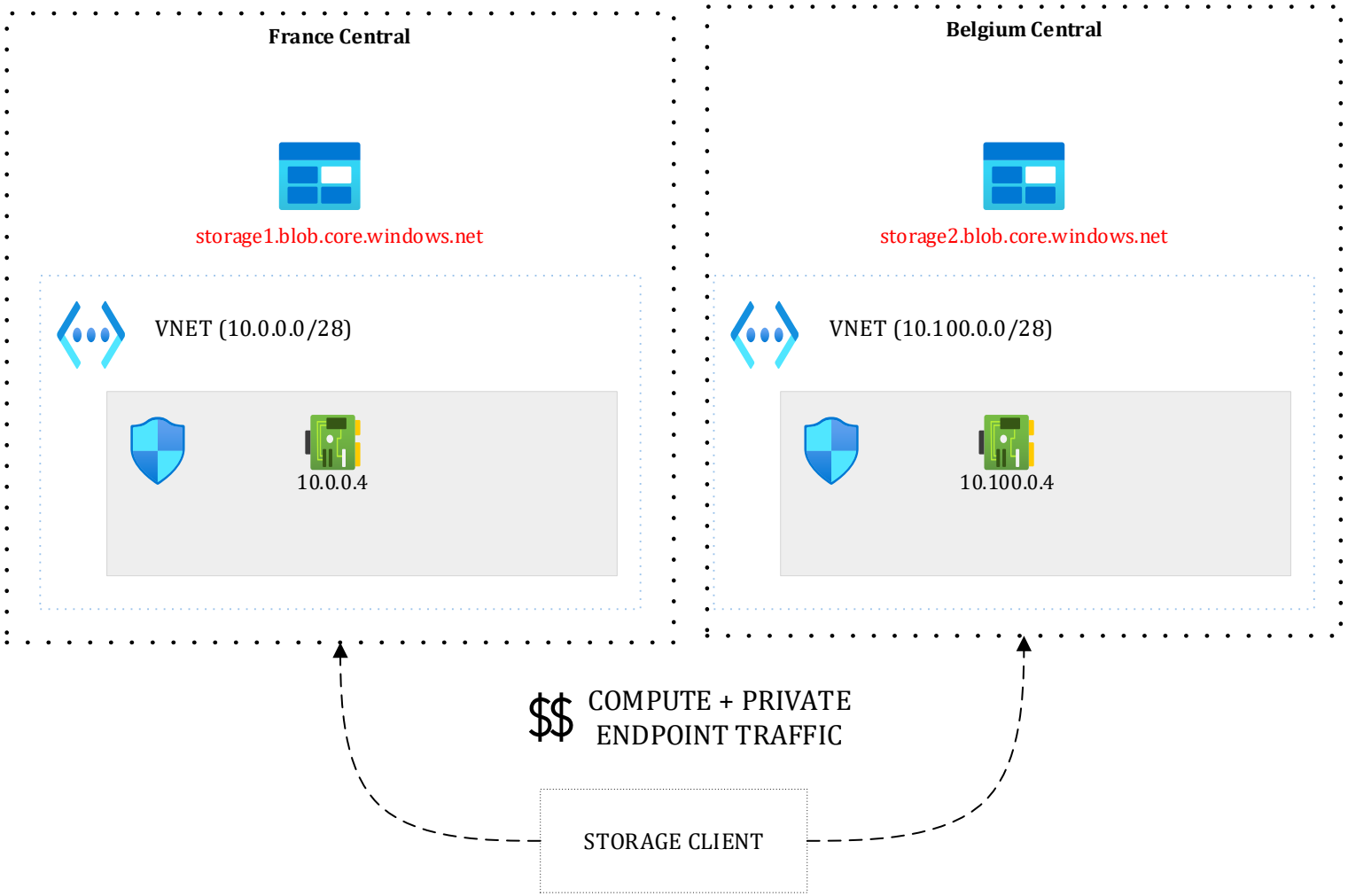
Double ingestion

Infra-level synchronization

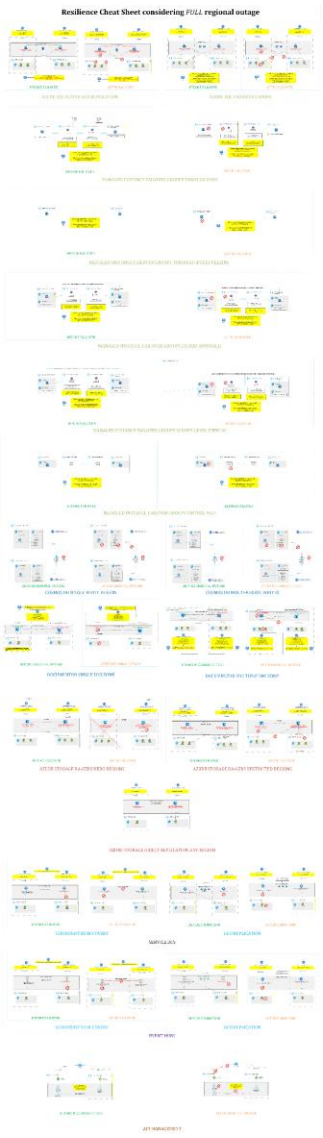
Object Replication – Any region



Double Data Ingestion or Infra-level Synchronization



Option	Region Types	Scope	Pros	Cons
GRS / RA-GRS / GZRS	Hero regions, Restricted regions (storage only)	Blobs, Queues, Tables, Files (with limits)	• Simple to configure • Managed by Azure • Automatic storage promotion on failover • Good availability during regional outage	• Asynchronous replication → potential data loss • Not available in local regions
GRS with external compute region	Restricted regions. Technically feasible in hero regions but would be silly.	Same as GRS	• Still benefits from managed replication • Satisfies long-distance regulatory requirements	• Cross-region latency after failover • More complex architecture • Compute and storage are no longer co-located
ZRS	All regions (recommended in restricted/local regions)	Same as LRS	• High availability within a region • No replication lag	• Does not survive full regional outage • No geographic isolation
Object Replication	All regions	Blobs only	• Cross-region replication without GRS • Explicit source/destination model • Works in local regions	• Read-only replicated containers • No automatic failover • Not supported for ADLS Gen2 (as of 01/2026)
Application-driven dual writes	All region types	Any storage type	• Maximum flexibility • Full control over consistency and failover logic	• Higher implementation complexity • Double ingestion and networking costs • Requires custom reconciliation logic
Job-based sync (AzCopy, etc.)	All region types	Mostly blobs/files	• Simple	• Asynchronous replication → potential data loss • Requires extra compute • Double ingestion costs • Operational overhead



Want more?

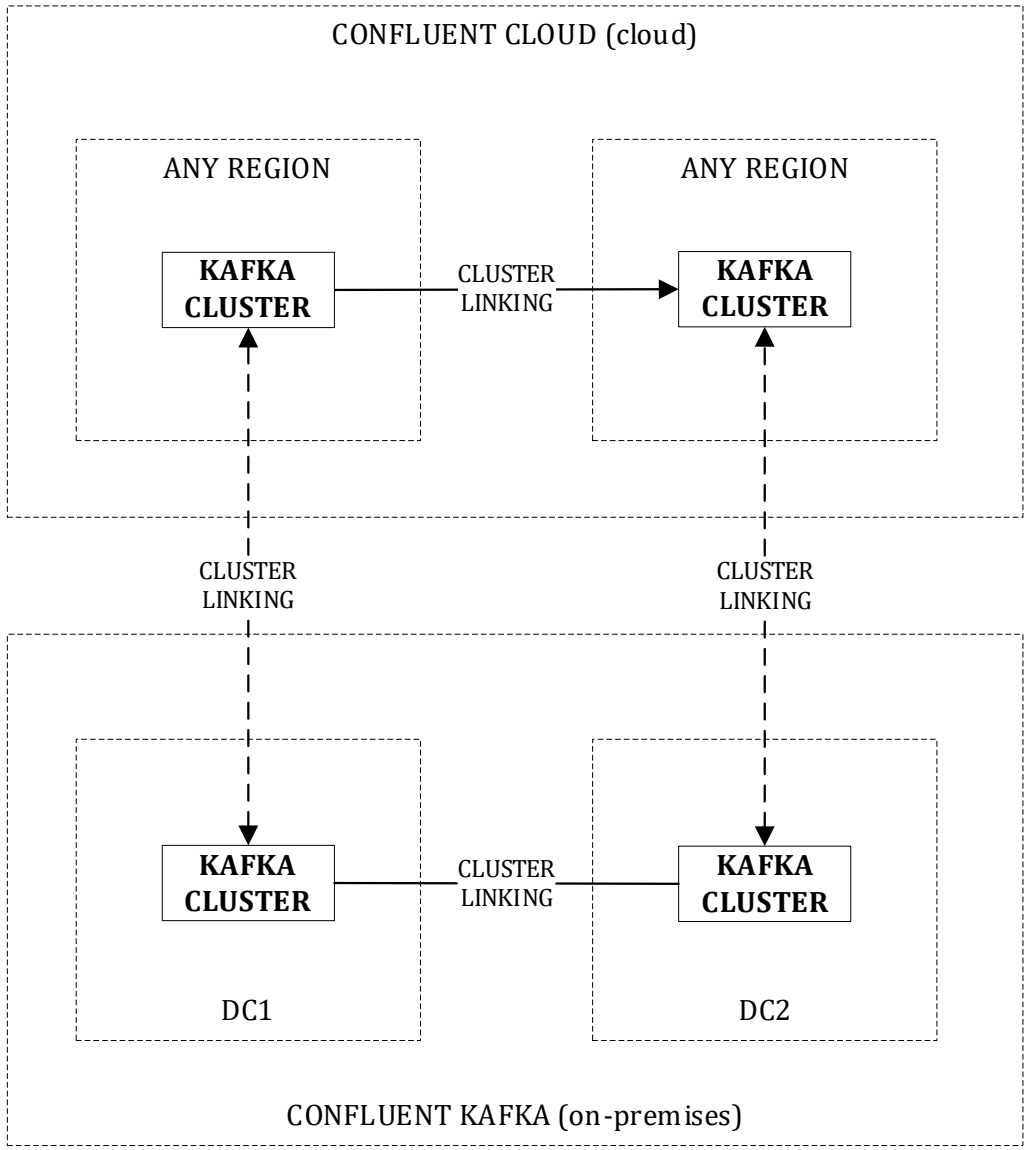
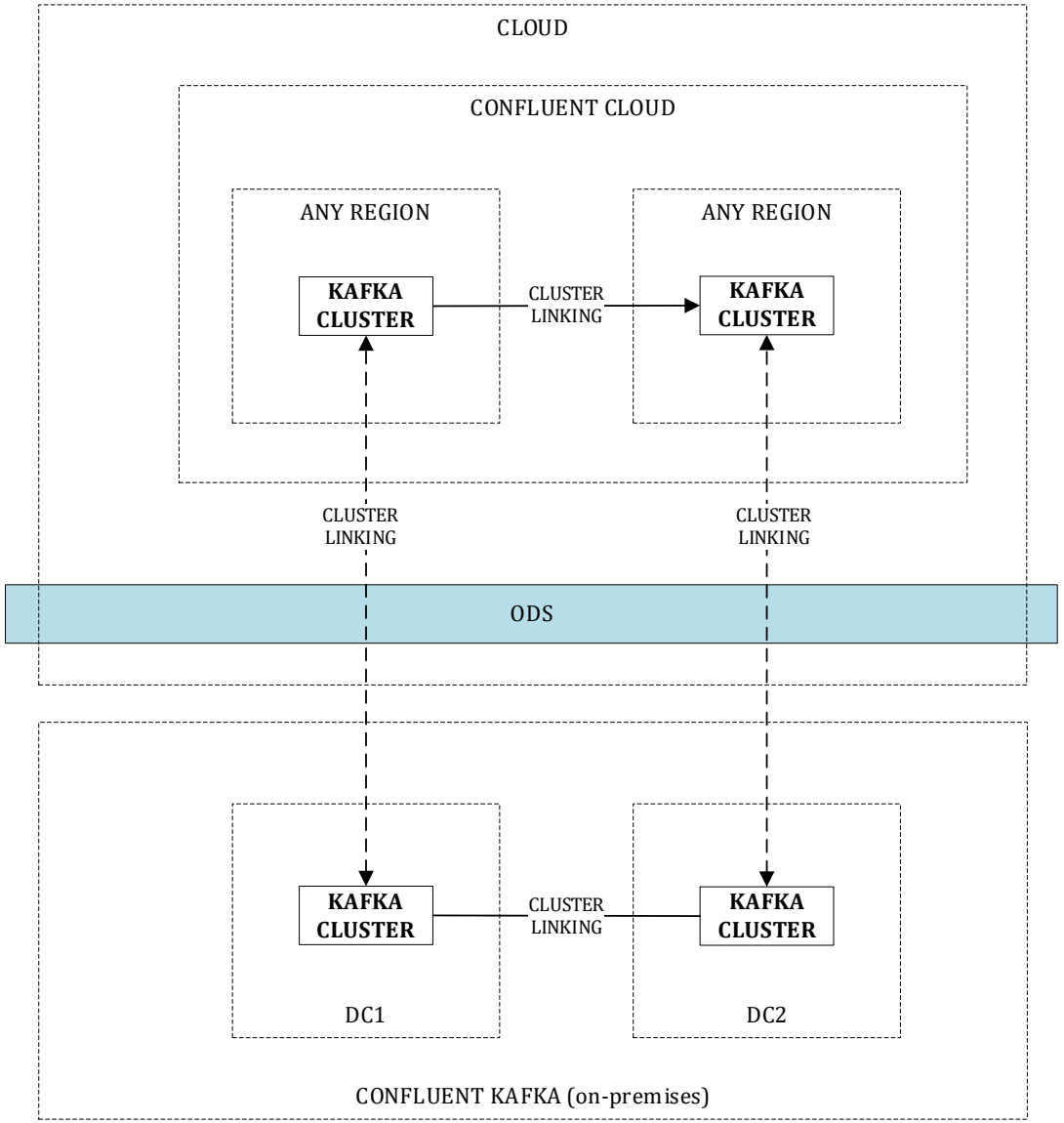
- https://github.com/stephaneey/azure-and-k8s-architecture/blob/main/cheat%20sheets/resilience.md

We only talked about Cloud only solutions and mostly about infrastructure building blocks but what about hybrid solutions?

Infrastructure + Application-level patterns in a hybrid world

- Network and DNS foundations (DNS GLB), Bow-Tie ExpressRoute design
- Cloud Infrastructure building blocks
- Application architecture styles may help achieve a better consistency and improve resilience:
  - Event-driven Architectures
    - Idempotency
    - Orchestrations
    - Outbox Pattern ➔ I will reliably publish events I produce
    - Inbox Pattern ➔ I will reliably process events I receive
    - Event sourcing ➔ Keeps a full history of all operations that happened and allows you to replay from any point in time.
  - Operational Data Stores (ODS)

Infrastructure+Application-level patterns in a hybrid world



Infrastructure+Application-level patterns